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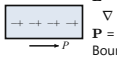
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1 · E, D, P: FIELDS IN DIELECTRICS

GENERAL SETUP



$\mathbf{E}$ = total field from ALL charge: $\nabla\cdot\mathbf{E} = \frac{\rho_f+\rho_b}{\epsilon_0}$, $\nabla\times\mathbf{E} = 0$, $\mathbf{E} = -\nabla V$

$\mathbf{P}$ = dipole moment per volume (C/m²)

Bound charge: $\sigma_b = \mathbf{P}\cdot\hat{n}$ ($\hat{n}$ out of material), $\rho_b = -\nabla\cdot\mathbf{P}$

$\mathbf{D} \equiv \epsilon_0\mathbf{E} + \mathbf{P}$ (C/m²): $\nabla\cdot\mathbf{D} = \rho_f$, $\oint \mathbf{D}\cdot d\mathbf{a} = Q_{f,enc}$

$\nabla\times\mathbf{D} = \nabla\times\mathbf{P}$ (NOT zero in general)

USEFUL FACTS

Linear: $\mathbf{P} = \epsilon_0\chi_e\mathbf{E}$ and $\mathbf{D} = \epsilon\mathbf{E}$

$\epsilon = \epsilon_0(1+\chi_e) = \epsilon_0\epsilon_r$, so $\epsilon_r = 1+\chi_e$, $\chi_e = \epsilon_r - 1$

Vacuum: $\chi_e = 0$, $\epsilon_r = 1$, $\epsilon = \epsilon_0$

$\mathbf{P} = \mathbf{D} - \epsilon_0\mathbf{E} = \frac{\chi_e}{\epsilon_r}\mathbf{D} = (1 - \frac{1}{\epsilon_r})\mathbf{D}$

$\mathbf{E} = \mathbf{D}/\epsilon$ (linear); $\mathbf{E} = (\mathbf{D} - \mathbf{P})/\epsilon_0$ (frozen P, no ε!)

Homogeneous linear: $\rho_b = -\frac{\chi_e}{1+\chi_e}\rho_f \Rightarrow$ no free charge inside ⇒ no ρ_b

Next to a conductor: $\sigma_f + \sigma_b = \sigma_f/\epsilon_r$

No free charge + symmetry ⇒ D = 0 ⇒ E = −P/ϵ_0 inside, 0 outside

Total bound charge always 0 (use as a check)

Recipe: D by Gauss (free only) → E = D/ε → P → σ_b, ρ_b → V from ∞ in

Typical ϵ_r: air 1.0005, glass 4–7, Si 12, water 80

SPECIFIC RESULTS

1.1 Uniformly polarized sphere



$\sigma_b = P\cos\theta$, $\rho_b = 0$

$\mathbf{E}_{in} = -\frac{\mathbf{P}}{3\epsilon_0}$ (uniform)

Outside: pure dipole $\mathbf{p} = \frac{4}{3}\pi R^3\mathbf{P}$

$V_{in} = \frac{P}{3\epsilon_0} r\cos\theta$, $V_{out} = \frac{PR^3\cos\theta}{3\epsilon_0 r^2}$

Why: = two balls ±ρ offset by d, ρd = P; E points + cap → − cap.

1.2 Ball with radial P = k r (HW 4.10)



$\rho_b = -3k$, $\sigma_b = kR$

$\mathbf{E}_{in} = -k\mathbf{r}/\epsilon_0$, $E_{out} = 0$

General $P = kr^n\hat{r}$: $\rho_b = -(n+2)kr^{n-1}$, $\sigma_b = kR^n$

Why: D = 0 by symmetry; bound charge totals 0.

1.3 Shell a<r<b with P = (k/r) r̂ (Gr 4.15)



$\rho_b = -k/r^2$

$\sigma_b(a) = -k/a$, $\sigma_b(b) = +k/b$

$E = -\frac{k}{\epsilon_0 r}\hat{r}$ in shell, 0 elsewhere

$P = \frac{k}{r^2}\hat{r}$: $\rho_b = 0$, $\sigma_b(a) = -\frac{k}{a^2}$, $\sigma_b(b) = \frac{k}{b^2}$

1.4 Long cylinder with P = k s ŝ



$\rho_b = -2k$, $\sigma_b = ka$, $E_{in} = -ks/\epsilon_0$

General $ks^n\hat{s}$: $\rho_b = -(n+1)ks^{n-1}$

Uniform P ⊥ axis: $\sigma_b = P\cos\phi$, $\mathbf{E}_{in} = -\mathbf{P}/2\epsilon_0$

1.5 Uniform P in a cylinder (bar electret) (HW 4.11)



$\sigma_b = \pm P$ on caps, 0 on sides

Long: $E_{in} \approx 0$; like point charges $\pm P\pi a^2$

Short disk: $E_{in} \approx -P/\epsilon_0$ (capacitor)

Shape factor: $\mathbf{E}_{in} = -N\mathbf{P}/\epsilon_0$

N: slab ⊥ 1, sphere ⅓, cylinder ⊥ ½, needle ∥ 0

1.6 Slab with P = P₀(z/a) ẑ, |z|<a



$\rho_b = -P_0/a$

$\sigma_b = +P_0$ on both faces

$E = -\frac{P_0 z}{a\epsilon_0}$ inside, 0 outside

1.7 Cavities and shells of uniform P

Uniformly polarized shell: **E = 0 in the cavity**

Spherical cavity in P-slab (P ⊥): $E = -\frac{2P}{3\epsilon_0}$

Outside that slab: dipole $-\frac{4}{3}\pi R^3\mathbf{P}$

1.8 Point charge at center of a dielectric ball



(charge −Q, susceptibility χ_e, radius R)

$\mathbf{D} = -\frac{Q}{4\pi r^2}\hat{r}$

$\mathbf{E} = -\frac{Q}{4\pi\epsilon_0(1+\chi_e)r^2}\hat{r}$

$\mathbf{P} = -\frac{\chi_e Q}{4\pi(1+\chi_e)r^2}\hat{r}$

$\sigma_b(R) = -\frac{\chi_e Q}{4\pi(1+\chi_e)R^2}$

$\rho_b = 0$ for r>0, plus point $+\frac{\chi_e Q}{1+\chi_e}$ at the origin

(from $\nabla\cdot\frac{\epsilon}{r^2} = 4\pi\delta^3$); outside $E = -\frac{Q}{4\pi\epsilon_0 r^2}$

1.9 Conductor (Q, radius a) + dielectric shell to b (HW 4.26)



$D = \frac{Q}{4\pi r^2}$, $E = \frac{Q}{4\pi\epsilon r^2}$ in shell

$V(a) = kQ[\frac{1}{b} + \frac{1}{\epsilon_r}(\frac{1}{a} - \frac{1}{b})]$

$\sigma_b(a) = -\frac{\chi_e}{\epsilon_r}\frac{Q}{4\pi a^2}$, $\sigma_b(b) = +\frac{\chi_e}{\epsilon_r}\frac{Q}{4\pi b^2}$

Two shells, at interface b: $\sigma_b = \frac{Q}{4\pi b^2}(\frac{1}{\epsilon_2} - \frac{1}{\epsilon_1})$

1.10 Coated wire λ / coax with dielectric



$D = \frac{\lambda}{2\pi s}$, $E = \frac{\lambda}{2\pi\epsilon s}$

$\sigma_b(a) = -\frac{\chi_e}{\epsilon_r}\frac{\lambda}{2\pi a}$, $\sigma_b(b) = +\frac{\chi_e}{\epsilon_r}\frac{\lambda}{2\pi b}$

$V(a) - V(b) = \frac{\lambda}{2\pi\epsilon}\ln\frac{b}{a}$

1.11 Ball of dielectric holding free ρ_f



$D = \frac{\rho_f r}{3}$ in, $\frac{Q}{4\pi r^2}$ out; $E_{in} = \frac{\rho_f r}{3\epsilon}$

$\rho_b = -\frac{\chi_e}{\epsilon_r}\rho_f$, $\sigma_b = \frac{\chi_e}{\epsilon_r}\frac{\rho_f R}{3}$

Free charge only on the surface ⇒ inside D = E = P = 0

1.12 Frozen P slab inside a capacitor (±σ)

$D = \sigma$ everywhere between plates

$E_{slab} = \frac{\sigma - P}{\epsilon_0}$, $E_{gap} = \frac{\sigma}{\epsilon_0}$

$\Delta V = \frac{\sigma d - P t}{\epsilon_0}$; $\sigma_b = \pm P$ on slab faces

2 · CAPACITORS & ENERGY

GENERAL SETUP



$C = Q/\Delta V$

$W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$

All charge: $W = \frac{\epsilon_0}{2}\int E^2 d\tau = \frac{1}{2}\int \rho V\, d\tau$

With dielectric (work on free charge): $W = \frac{1}{2}\int \mathbf{D}\cdot\mathbf{E}\, d\tau = \frac{1}{2}\int \frac{D^2}{\epsilon} d\tau$

USEFUL FACTS

Stacked layers = series (same D): $\frac{1}{C} = \sum\frac{1}{C_i}$

Side by side = parallel (same E): $C = \sum C_i$

Fill with ϵ_r: fixed Q ⇒ W ÷ ϵ_r; fixed V ⇒ W × ϵ_r

Force: $F = \frac{1}{2}V^2\frac{dC}{dx}$, toward higher C (fixed Q gives same F)

Conductor surface pressure: $\frac{\sigma^2}{2\epsilon_0}$

Outside a sphere the energy is always $\frac{Q^2}{8\pi\epsilon_0 R}$

Infinite line/cylinder: energy per length diverges outside

SPECIFIC RESULTS

2.1 Parallel plates, stacked slabs (HW 4.18)



$D = \sigma$ in both; $E_i = \sigma/\epsilon_i$; $P_i = \sigma(1 - \frac{1}{\epsilon_r})$

Interface: $\sigma_b = \sigma(\frac{1}{\epsilon_{r2}} - \frac{1}{\epsilon_{r1}})$

$\frac{1}{C} = \sum\frac{d_i}{\epsilon_i A}$; half-filled: $C = \frac{2\epsilon_r}{1+\epsilon_r}C_0$

4.18 (ε, 2, 1.5; thickness a): $V = \frac{7\sigma a}{6\epsilon_0}$

2.2 Parallel plates, side-by-side slabs (Disc 4.19)



$E = V/d$ in both; $\sigma_{f,i} = D_i = \epsilon_i E$

$C = \sum\frac{\epsilon_i A_i}{d}$; half-filled: $C = \frac{1+\epsilon_r}{2}C_0$

2.3 Coax (HW 4.21)



$\frac{C}{L} = \frac{2\pi\epsilon_0}{\sum\ln(b_i/a_i)/\epsilon_{ri}}$ (layers in series)

Half-filled by angle: $\frac{C}{L} = \frac{\pi\epsilon_0(1+\epsilon_r)}{\ln(b/a)}$

2.4 Spherical capacitor



$C = \frac{4\pi\epsilon_0 ab}{b-a}$; isolated sphere $4\pi\epsilon_0 R$

Layers: $\frac{1}{C} = \sum\frac{1}{4\pi\epsilon_i}(\frac{1}{a_i} - \frac{1}{b_i})$

$\epsilon = \epsilon_0\frac{b}{r}$: $C = \frac{4\pi\epsilon_0 ab}{\ln(b/a)}$

2.5 Permittivity that varies



$D = \sigma$; $E = \sigma/\epsilon(x)$; $C = A/\int_0^d \frac{dx}{\epsilon}$; $\rho_b = -dP/dx$

$\epsilon = \epsilon_0(1 + \frac{x}{d})$: $C = \frac{\epsilon_0 A}{d\ln 2}$

Linear $\epsilon_1 \to \epsilon_2$: $C = \frac{A(\epsilon_2 - \epsilon_1)}{d\ln(\epsilon_2/\epsilon_1)}$

2.6 Dielectric pulled into a capacitor



Width w, gap d, inserted x: $C = \frac{\epsilon_0 w}{d}(\ell + \chi_e x)$

$F = \frac{\epsilon_0\chi_e w V^2}{2d}$ (pulled in)

2.7 Oil rising in a coax (Gr 4.28)



$F = \frac{\pi\epsilon_0\chi_e V^2}{\ln(b/a)}$

$h = \frac{\epsilon_0\chi_e V^2}{\rho g(b^2-a^2)\ln(b/a)}$

2.8 Standard energies



Shell $\frac{Q^2}{8\pi\epsilon_0 R}$; uniform ball $\frac{3Q^2}{20\pi\epsilon_0 R}$

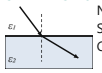
Ball of ε, with ρ_f: $\frac{Q^2}{8\pi\epsilon_0 R}(1 + \frac{1}{5\epsilon_r})$

Conductor + shell: $\frac{Q^2}{8\pi\epsilon_0\epsilon_r}(\frac{1}{a} + \frac{\chi_e}{b})$

Coax per length: $\frac{\lambda^2}{4\pi\epsilon}\ln\frac{b}{a}$

3 · BOUNDARY CONDITIONS & LAPLACE PROBLEMS

GENERAL SETUP



No charge in a region ⇒ $\nabla^2 V = 0$

Sphere: $V = \sum(Ar^l + \frac{B_l}{r^{l+1}})P_l(\cos\theta)$

Cylinder: $A_0 + B_0\ln s + \sum(A_n s^n + B_n s^{-n})(C_n\cos n\phi + D_n\sin n\phi)$

Box: $e^{\pm kx}$ sin ky, sinh, cosh

$D_1^\perp - D_2^\perp = \sigma_f$; $\mathbf{E}^\parallel$ continuous; V continuous

No free σ: $\epsilon_1\partial_n V_1 = \epsilon_2\partial_n V_2$

Charged sheet in vacuum: $\partial_n V_{out} - \partial_n V_{in} = -\sigma/\epsilon_0$

USEFUL FACTS

Center finite ⇒ drop $r^{-(l+1)}$, s^{-n}, ln s

Far away ⇒ drop r^l, except $-E_0 r\cos\theta$ for a uniform field

Uniform field ⇒ only l = 1 (n = 1) terms appear

$P_0 = 1$, $P_1 = \cos\theta$, $P_2 = \frac{3\cos^2\theta - 1}{2}$, $P_3 = \frac{5\cos^3\theta - 3\cos\theta}{2}$

$\cos^2\theta = \frac{1}{3} + \frac{2}{3}P_2$; $c_l = \frac{2l+1}{2}\int_{-1}^{1} f(x)P_l(x)dx$

Field lines bend: $\frac{\tan\theta_1}{\tan\theta_2} = \frac{\epsilon_1}{\epsilon_2}$ (away from normal in high ε)

Uniqueness: anything that meets every BC is THE answer

Check every answer: $\epsilon_r \to 1$ (vacuum), $\epsilon_r \to \infty$ (conductor)

SPECIFIC RESULTS

3.1 Dielectric sphere in uniform E_0



$V_{in} = Ar\cos\theta$; $V_{out} = -E_0 r\cos\theta + \frac{B\cos\theta}{r^2}$

BCs at R: $AR = -E_0 R + \frac{B}{R^2}$ and $\epsilon_r A = -E_0 - \frac{2B}{R^3}$

$E_{in} = \frac{3E_0}{\epsilon_r+2}$ uniform; $B = \frac{\epsilon_r-1}{\epsilon_r+2}E_0 R^3$

$P = 3\epsilon_0\frac{\epsilon_r-1}{\epsilon_r+2}E_0$; $\sigma_b = P\cos\theta$

Just outside the pole: $E = \frac{3\epsilon_r}{\epsilon_r+2}E_0$

3.2 Conducting sphere in uniform E_0

$V = -E_0(r - \frac{R^3}{r^2})\cos\theta$

$\sigma = 3\epsilon_0 E_0\cos\theta$; induced $p = 4\pi\epsilon_0 R^3 E_0$

With net charge Q: add $\frac{kQ}{r}$

Conducting cylinder ⊥: $V = -E_0(s - \frac{R^2}{s})\cos\phi$, $\sigma = 2\epsilon_0 E_0\cos\phi$

3.3 Bubble, or any ϵ_{in} inside ϵ_{out}



Sphere: $E_{in} = \frac{3\epsilon_{out}}{\epsilon_{in}+2\epsilon_{out}}E_0$

Air cavity in ϵ_r: $\frac{3\epsilon_r}{1+2\epsilon_r}E_0$ (stronger than E_0)

Cylinder ⊥: $\frac{2\epsilon_{out}}{\epsilon_{in}+\epsilon_{out}}E_0$; dielectric cyl: $\frac{2E_0}{\epsilon_r+1}$

3.4 Coated conductor in uniform E_0 (HW 4.24)



Shell: $V = A(r - \frac{a^3}{r^2})\cos\theta$

$A = \frac{-3E_0}{(\epsilon_r+2)+2(\epsilon_r-1)a^3/b^3}$

$\mathbf{E} = -A[(1 + \frac{2a^3}{r^3})\cos\theta\hat{r} - (1 - \frac{a^3}{r^3})\sin\theta\hat{\theta}]$

3.5 Spherical shell with V(θ) or σ(θ)



Given $V = \sum c_l P_l$: in $c_l(\frac{r}{R})^l$, out $c_l(\frac{R}{r})^{l+1}$

$\sigma = \frac{\epsilon_0}{R}\sum(2l+1)c_l P_l$

Given $\sigma = \sum\sigma_l P_l$: $A_l = \frac{\sigma_l}{(2l+1)\epsilon_0 R^{l-1}}$, $B_l = A_l R^{2l+1}$

$\sigma = k\cos\theta$: $E_{in} = \frac{k}{3\epsilon_0}$, pointing down

3.6 Slot / box (Cartesian)



Two zero walls ⇒ $\sin\frac{n\pi y}{a}$; other way $e^{-n\pi x/a}$, sinh, cosh

Constant V_0 on the end: $C_n = \frac{4V_0}{n\pi}$, n odd

$V = \frac{4V_0}{\pi}\sum_{odd} \frac{1}{n}e^{-n\pi x/a}\sin\frac{n\pi y}{a}$

3.7 Wedge / pipe (polar)



Walls at 0, α ⇒ $r^{\pm k}\sin k\theta$, $k = \frac{n\pi}{\alpha}$

Zero arc at a: $(\frac{r}{a})^k - (\frac{a}{r})^k$; one wall at V_0: $V = \frac{V_0\theta}{\alpha}$

4 · IMAGES

GENERAL SETUP



Replace a conductor (or dielectric) by fictitious charges placed OUTSIDE the region you solve in

If V matches on every boundary and at ∞, uniqueness ⇒ correct there

USEFUL FACTS

Induced charge: $\sigma = -\epsilon_0\partial V/\partial n$; total induced = sum of images

Force on q: from the images only

Energy to bring q in: ½ × q × (V of images)

SPECIFIC RESULTS

4.1 Charge above a grounded plane



Image −q at the mirror point

$F = \frac{kq^2}{4d^2}$ (pull); $\sigma = -\frac{qd}{2\pi(a^2+d^2)^{3/2}}$; $W = -\frac{kq^2}{4d}$

Plane held at V_0: add V_0

4.2 Charge near a grounded sphere



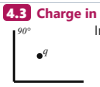
$q' = -\frac{qR}{a}$ at $\frac{R^2}{a}$; $F = \frac{kq^2 Ra}{(a^2-R^2)^2}$

Neutral sphere: add $+\frac{qR}{a}$ at center

Sphere at V_0: add $4\pi\epsilon_0 RV_0$ at center

q inside a grounded shell at d: image $-\frac{qR}{d}$ at $\frac{R^2}{d}$

4.3 Charge in a 90° corner



Images: $-q$ at $(-x_0, y_0)$, $-q$ at $(x_0, -y_0)$, $+q$ at $(-x_0, -y_0)$

4.4 Charge above a dielectric half-space



Above: image $q' = -\frac{\epsilon_r-1}{\epsilon_r+1}q$ at −d

Below: $q'' = \frac{2q}{\epsilon_r+1}$ at +d

BCs: $q + q' = q''$ and $q - q' = \epsilon_r q''$

$F = \frac{kqq'}{(2d)^2}$; $\sigma_b = -\frac{1}{2\pi}\frac{\chi_e}{\chi_e+2}\frac{qd}{(a^2+d^2)^{3/2}}$

4.5 Dipole above a grounded plane (HW 4.6)



Image: component ∥ plane flips, ⊥ component kept

$N = \frac{kp^2\sin\theta\cos\theta}{(2z)^3}$; rests ⊥ to plane

4.6 Line charge above a plane



Image $-\lambda$ (dielectric: $\times\frac{\epsilon_r-1}{\epsilon_r+1}$)

$V = \frac{\lambda}{2\pi\epsilon_0}\ln\frac{s_-}{s_+}$; force/length $\frac{\lambda^2}{4\pi\epsilon_0 d}$

5 · MULTIPOLES & DIPOLES

GENERAL SETUP



$V = k\sum_n \frac{1}{r^{n+1}}\int r'^n P_n(\cos\alpha)\rho\, d\tau'$

$= k[\frac{Q}{r} + \frac{\mathbf{p}\cdot\hat{r}}{r^2} + \frac{1}{r^3}\sum q_i r_i^2 P_2(\cos\alpha_i) + \dots]$

$Q = \int\rho\, d\tau$; $\mathbf{p} = \int\mathbf{r}'\rho\, d\tau' = \sum q_i\mathbf{r}_i$

USEFUL FACTS

Far away the first nonzero moment wins: $V \propto r^{-(n+1)}$, $E \propto r^{-(n+2)}$

Shift origin by $\mathbf{a}$: $\mathbf{p}' = \mathbf{p} - Q\mathbf{a}$ (Q = 0 ⇒ p independent of origin)

Dipole in a field: $\mathbf{N} = \mathbf{p}\times\mathbf{E}$, $U = -\mathbf{p}\cdot\mathbf{E}$, $\mathbf{F} = \nabla(\mathbf{p}\cdot\mathbf{E})$

Uniform E: F = 0; aligned dipole is pulled toward stronger E

SPECIFIC RESULTS

5.1 Field of a dipole

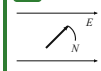


$V = \frac{kp\cos\theta}{r^2}$

$\mathbf{E} = \frac{kp}{r^3}(2\cos\theta\hat{r} + \sin\theta\hat{\theta}) = \frac{k}{r^5}[3(\mathbf{p}\cdot\hat{r})\hat{r} - \mathbf{p}]$

On axis $\frac{2kp}{r^3}$; on equator $-\frac{kp}{r^3}$

5.2 Forces between dipoles and charges (HW 4.7)



Coaxial dipoles: $F = \frac{3p_1p_2}{2\pi\epsilon_0 r^4}$ (attract)

$U_{12} = \frac{k}{r^3}[\mathbf{p}_1\cdot\mathbf{p}_2 - 3(\mathbf{p}_1\cdot\hat{r})(\mathbf{p}_2\cdot\hat{r})]$

Aligned dipole near charge q: $F = \frac{2kqp}{r^3}$

Sphere near q (r ≫ R): $\mathbf{p} = 4\pi\epsilon_0 R^3\frac{\epsilon_r-1}{\epsilon_r+2}\mathbf{E}$, $F \propto \frac{1}{r^5}$

5.3 Dipole moments to know



±q separated by d: $p = qd$ (from − to +)

Hemisphere shell σ: $\pi\sigma R^3$; solid hemisphere ρ: $\frac{\pi\rho R^4}{4}$

Shell $k\cos\theta$: $\frac{4}{3}\pi kR^3$; ball $k\cos\theta$: $\frac{\pi kR^4}{3}$

Ring $\lambda_0\cos\phi$: $\pi R^2\lambda_0$; rod $\lambda_0 z/L$ on [−L, L]: $\frac{2}{3}\lambda_0 L^2$

5.4 Linear quadrupole (q, −2q, q)



Q = 0, p = 0: $V = \frac{kqa^2(3\cos^2\theta-1)}{r^3}$

Force in nonuniform E: add $q_i\mathbf{E}(\mathbf{r}_i)$

5.5 Overlapping balls ±ρ



Overlap: $\mathbf{E} = -\frac{\rho\mathbf{d}}{3\epsilon_0}$ uniform (d from − center to + center)

Cylinders: $-\frac{\rho\mathbf{d}}{2\epsilon_0}$; d → 0, ρd = P: polarized sphere

6 · EXAM I BASICS: COULOMB, GAUSS, CONDUCTORS

GENERAL SETUP

$\mathbf{E} = k\int\frac{dq}{|\mathbf{s}-\mathbf{r}'|^2}\hat{}$, $V = k\int\frac{dq}{|\mathbf{s}-\mathbf{r}'|}$, $dq = \lambda dl$, σda, $\rho d\tau$

$\oint\mathbf{E}\cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0}$; $\nabla\cdot\mathbf{E} = \frac{\rho}{\epsilon_0}$; $\nabla^2 V = -\frac{\rho}{\epsilon_0}$

$V(b) - V(a) = -\int_a^b\mathbf{E}\cdot d\mathbf{l}$; $W = q\Delta V$

USEFUL FACTS

Conductor: E = 0 inside, V constant, charge on surface, $E = \sigma/\epsilon_0$ just outside

Charge q in a cavity ⇒ −q on the cavity wall

$E^\perp$ jumps by σ/ϵ_0; $E^\parallel$ and V continuous

Given E: $\rho = \epsilon_0\nabla\cdot\mathbf{E}$; $Q_{enc}(r) = 4\pi\epsilon_0 r^2 E_r$; ≠ 0 as r → 0 ⇒ point charge

$\nabla\cdot\frac{\hat{r}}{r^2} = 4\pi\delta^3(\mathbf{r})$; $\nabla\cdot\frac{\hat{s}}{s} = 2\pi\delta^2$

SPECIFIC RESULTS

6.1 Point charge / ball



Outside: $\frac{kQ}{r^2}$; inside uniform ball: $\frac{\rho r}{3\epsilon_0}$

$V_{in} = \frac{kQ}{2R}(3 - \frac{r^2}{R^2})$; $\rho = kr^n$: $E = \frac{kr^{n+1}}{(n+3)\epsilon_0}$

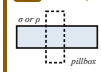
6.2 Line / cylinder



Line: $\frac{\lambda}{2\pi\epsilon_0 s}$, $V = -\frac{\lambda}{2\pi\epsilon_0}\ln\frac{s}{s_0}$

Inside uniform cylinder: $\frac{\rho s}{2\epsilon_0}$; $\rho = ks^n$: $\frac{ks^{n+1}}{(n+2)\epsilon_0}$

6.3 Sheet / slab



Sheet: $\frac{\sigma}{2\epsilon_0}$ each side; conductor surface $\frac{\sigma}{\epsilon_0}$

Slab |z|<a: $\frac{\rho z}{\epsilon_0}$ inside, $\frac{\rho a}{\epsilon_0}$ outside (constant)

6.4 Ring / disk / segment (axis)



Ring: $E = \frac{kQz}{(z^2+R^2)^{3/2}}$, $V = \frac{kQ}{\sqrt{z^2+R^2}}$

Disk: $E = \frac{\sigma}{2\epsilon_0}(1 - \frac{z}{\sqrt{z^2+R^2}})$

Segment: $E_\perp = \frac{k\lambda}{s}(\sin\theta_2 - \sin\theta_1)$

6.5 Overlaps and cavities



Hole = whole shape + negative copy; each field from its own center

Cavity offset d in ball ρ: $\mathbf{E} = \frac{\rho\mathbf{d}}{3\epsilon_0}$ uniform

7 · MATH

Div/curl one-liners

$F(r)\hat{r}$: div $\frac{1}{r^2}(r^2F)'$

$F(s)\hat{s}$: div $\frac{1}{s}(sF)'$

$F(s)\hat{\phi}$: curl $\frac{1}{s}(sF)'\hat{z}$

$F(s)\hat{z}$: curl $-F'\hat{\phi}$

$F(z)\hat{x}$: curl $F'\hat{y}$

Cylindrical

$\nabla\cdot\mathbf{A} = \frac{1}{s}\partial_s(sA_s) + \frac{1}{s}\partial_\phi A_\phi + \partial_z A_z$

$\nabla\times\mathbf{A} = [\frac{1}{s}\partial_\phi A_z - \partial_z A_\phi]\hat{s} + [\partial_z A_s - \partial_s A_z]\hat{\phi} + \frac{1}{s}[\partial_s(sA_\phi) - \partial_\phi A_s]\hat{z}$

$\nabla^2 f = \frac{1}{s}\partial_s(s\partial_s f) + \frac{1}{s^2}\partial_\phi^2 f + \partial_z^2 f$

Spherical

$\nabla\cdot\mathbf{A} = \frac{1}{r^2}\partial_r(r^2A_r) + \frac{\partial_\theta(\sin\theta A_\theta)}{r\sin\theta} + \frac{\partial_\phi A_\phi}{r\sin\theta}$

$\nabla^2 f = \frac{1}{r^2}\partial_r(r^2\partial_r f) + \frac{\partial_\theta(\sin\theta\partial_\theta f)}{r^2\sin\theta}$

Unit vectors

$\hat{s} = \cos\theta\hat{r} - \sin\theta\hat{\theta}$; adjoint $\hat{\theta} = -\hat{s}$

$\hat{s} = \cos\phi\hat{x} + \sin\phi\hat{y}$; $\hat{\phi} = -\sin\phi\hat{x} + \cos\phi\hat{y}$

$\hat{s}\times\hat{\phi} = \hat{z}$, $\hat{\phi}\times\hat{z} = \hat{s}$, $\hat{z}\times\hat{s} = \hat{\phi}$, $\hat{r}\times\hat{\theta} = \hat{\phi}$

Integrals

$\int\frac{dz}{(z^2+s^2)^{3/2}} = \frac{z}{s^2\sqrt{z^2+s^2}}$

$\int\frac{z\,dz}{(z^2+s^2)^{3/2}} = -\frac{1}{\sqrt{z^2+s^2}}$

$\int\frac{dz}{\sqrt{z^2+s^2}} = \ln(z + \sqrt{z^2+s^2})$

$\int_0^\pi\sin^3\theta\, d\theta = \frac{4}{3}$; $\int_0^\pi\cos^2\theta\sin\theta\, d\theta = \frac{2}{3}$

$\int_0^\infty x^n e^{-x/a}dx = n!\,a^{n+1}$

$\int_{-1}^{1}P_l^2 dx = \frac{2}{2l+1}$; $\int_0^1 P_l\, dx = 1, \frac{1}{2}, 0, -\frac{1}{8}$

Units

P, D, σ: C/m²; E: V/m; p: C·m; $\epsilon_0 = 8.85\times10^{-12}$; $k = 8.99\times10^9$

PHYS 435 · Exam II · BACK: MAGNETIC (sections 8–13)

$\mu_0 = 4\pi \times 10^{-7}$; RHR: thumb along I, fingers curl with B

CONTENTS: find your setup, go to its number

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8.2 Arcs + straight leads (hairpin)
8.3 Circular loop, on its axis
8.4 Square / polygon / rectangle, center
8.5 Finite solenoid, on axis
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10.6 Given A: find B, then J

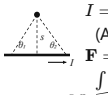
11 MAGNETIC DIPOLES & SPINNING CHARGE
11: setup, facts
11.1 Two dipoles
11.2 Spinning disk (σ, ω)
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12 B, H, M: MAGNETIC MATTER
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12.1 Uniformly magnetized sphere
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12.3 Cylinder with M along φ̂
12.4 Cylinder with M = k s ẑ
12.5 Linear slab carrying free J_0 x̂, |z| <d/2
12.6 Coax / toroid / solenoid with χ_m
12.7 Linear sphere / slab in uniform B_0

13 MAGNETIC BOUNDARY CONDITIONS & E ↔ B
13: setup, facts

8 · CURRENTS, FORCES, BIOT–SAVART

GENERAL SETUP



$I = \lambda v$ (A); $\mathbf{K} = \sigma \mathbf{v}$ (A/m); $\mathbf{J} = \rho \mathbf{v}$ (A/m²)

$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$; on a wire $\mathbf{F} = \int I\, d\mathbf{l} \times \mathbf{B}$

$\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{I\, d\mathbf{l} \times \hat{\mathbf{r}}}{|\mathbf{r}-\mathbf{r}'|^2}$ (or $\mathbf{K}\, da$, $\mathbf{J}\, d\tau$)

Steady currents: $\nabla \cdot \mathbf{J} = 0$

USEFUL FACTS

RHR: thumb along I, fingers curl with B

Pieces: straight piece whose LINE passes through P gives 0

Arc of angle φ, radius R, at center: $\frac{\mu_0 I \phi}{4\pi R}$

Semi-infinite wire ending beside P (distance s): $\frac{\mu_0 I}{4\pi s}$

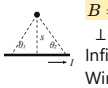
Spinning: disk $K = \sigma\omega s$; shell $K = \sigma\omega R \sin\theta$; ball $J = \rho\omega r \sin\theta$; ring $I = \frac{Q\omega}{2\pi}$

Uniform B: closed loop F = 0, torque $\mathbf{m} \times \mathbf{B}$; curved wire = straight chord

Cyclotron: $r = \frac{mv}{qB}$, $\omega = \frac{qB}{m}$; B does no work

SPECIFIC RESULTS

8.1 Straight segment



$B = \frac{\mu_0 I}{4\pi s}(\sin\theta_2 - \sin\theta_1)$ (θ from the ⊥ foot)

Infinite: $\frac{\mu_0 I}{2\pi s}$

Wire ∥ z at (x_0, y_0): $\mathbf{B} = \frac{\mu_0 I}{2\pi} \frac{(-(y-y_0), x-x_0, 0)}{(x-x_0)^2+(y-y_0)^2}$

8.2 Arcs + straight leads (hairpin)



Loop center $\frac{\mu_0 I}{2R}$; half loop $\frac{\mu_0 I}{4R}$

Hairpin: $\frac{\mu_0 I}{4R} + \frac{\mu_0 I}{2\pi R}$

Two arcs a, b + radial: $\frac{\mu_0 I \phi}{4\pi}(\frac{1}{a} - \frac{1}{b})$

8.3 Circular loop, on its axis HW 5.37



$B = \frac{\mu_0 I R^2}{2(R^2+z^2)^{3/2}}$

Far: $\frac{\mu_0 I R^2}{2z^3}$ (dipole $m = I\pi R^2$); Helmholtz d = R: $\frac{8\mu_0 I}{5\sqrt{5}R}$

8.4 Square / polygon / rectangle, center HW 5.8



Square (center to side R): $\frac{\sqrt{2}\mu_0 I}{\pi R}$

n-gon: $\frac{n\mu_0 I}{2\pi R} \sin\frac{\pi}{n}$; rectangle: $\frac{2\mu_0 I\sqrt{L_1^2+L_2^2}}{\pi L_1 L_2}$

8.5 Finite solenoid, on its axis Disc 5.11



$B = \frac{\mu_0 n I}{2}(\cos\theta_2 - \cos\theta_1)$

Mouth of a semi-infinite one: $\frac{\mu_0 n I}{2}$

8.6 Forces between currents HW 5.10, Disc 5.4



Parallel wires: $f = \frac{\mu_0 I_1 I_2}{2\pi d}$ (same direction attract)

Square loop near wire: $\frac{\mu_0 I^2 a^2}{2\pi s(s+a)}$ toward the wire

$\mathbf{B} = kz\hat{x}$, square loop in yz: $\mathbf{F} = Ika^2\hat{z}$

9 · AMPÈRE & SUPERPOSITION

GENERAL SETUP



$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$

$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$; $\nabla \cdot \mathbf{B} = 0$

USEFUL FACTS

Works only for line, plane, solenoid, toroid symmetry (or sums)

Symmetry: $B_s = 0$ (∇·B = 0), $B_z = 0$ (rectangle to ∞) ⇒ $\mathbf{B} = B(s)\hat{\phi}$

Loops: circle (wire), rectangle straddling (sheet), rectangle half inside (solenoid)

$I_{enc} = \int J(s)\, 2\pi s\, ds$

Inside a uniform wire: $\mathbf{B} = \frac{\mu_0}{2}\mathbf{J} \times \mathbf{s}$ (s from that wire's axis)

Holes/overlaps: superpose full shapes with +J and −J

SPECIFIC RESULTS

9.1 Solid wire, uniform J



Inside $\frac{\mu_0 I s}{2\pi a^2}$; outside $\frac{\mu_0 I}{2\pi s}$

Current only on surface: 0 inside

9.2 Wire with J ∝ s^n; thick tube HW 5.14



$I_{enc} = I(\frac{s}{a})^{n+2}$, $B = \frac{\mu_0 I}{2\pi s}(\frac{s}{a})^{n+2}$

$J = ks$: $B_{in} = \frac{\mu_0 I s^2}{2\pi a^3}$; $J = k/s$: $B = \mu_0 k$

Tube a..b: $\frac{\mu_0 I}{2\pi s}\frac{s^2-a^2}{b^2-a^2}$

9.3 Coax



Between: $\frac{\mu_0 I}{2\pi s}$; outside: 0

9.4 Infinite current sheet K



$\mathbf{B} = \frac{\mu_0}{2}\mathbf{K} \times \hat{n}$ ($\hat{n}$ from sheet toward point)

Same size both sides, opposite directions, no falloff

9.5 Moving capacitor (two opposite sheets) HW 5.18



K = σv on each plate, in opposite directions

Between: $B = \mu_0 \sigma v$; outside 0

Direction: $\mathbf{K}_{top} \times$(unit vector from top plate into gap)

+σ at y = d, −σ at y = 0, v x̂: $\mathbf{B} = -\mu_0 \sigma v \hat{z}$

Force/area on top: $\frac{\mu_0 \sigma^2 v^2}{2}$ (repel) = $\frac{\sigma^2}{2\epsilon_0}$ at v = c

9.6 Thick slab with J x̂, |z|<a Disc 5.15



$\mathbf{B} = -\mu_0 J z \hat{y}$ inside; $\mp\mu_0 J a \hat{y}$ above/below

9.7 Solenoid / toroid HW 5.17



Solenoid $\mu_0 nI$ inside, 0 outside; toroid $\frac{\mu_0 NI}{2\pi s}$

Coaxial, opposite: $\mu_0(n_1 - n_2)I$, $\mu_0 n_2 I$, 0

9.8 Cylinder with an off-axis hole



Wire radius a, hole radius c, hole axis at $\mathbf{b} = b\hat{x}$, total I (+z)

$J = \frac{I}{\pi(a^2-c^2)}$; $I_+ = \frac{Ia^2}{a^2-c^2}$, $I_- = -\frac{Ic^2}{a^2-c^2}$

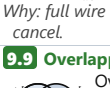
In hole: $\mathbf{B} = \frac{\mu_0}{2}\mathbf{J} \times \mathbf{b} = \frac{\mu_0 Ib}{2\pi(a^2-c^2)}\hat{y}$ (uniform)

In metal: $\frac{\mu_0}{2}\mathbf{J} \times \mathbf{s} + \frac{\mu_0 I_-}{2\pi s'}\hat{\phi}'$

Outside: $\frac{\mu_0 I_+}{2\pi s}\hat{\phi} + \frac{\mu_0 I_-}{2\pi s'}\hat{\phi}'$

Why: full wire (+J) + hole wire (−J); positions cancel.

9.9 Overlapping wires ±J; slab with hole



Overlap: $\mathbf{B} = \frac{\mu_0}{2}\mathbf{J} \times (\mathbf{c}_- - \mathbf{c}_+)$ uniform

Slab $J\hat{x}$ with hole along x: $-\frac{\mu_0 J}{2}(z\hat{y} + y\hat{z})$

10 · VECTOR POTENTIAL A

GENERAL SETUP



$\mathbf{B} = \nabla \times \mathbf{A}$; Coulomb gauge $\nabla \cdot \mathbf{A} = 0$

$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$; $\mathbf{A} = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}}{|\mathbf{r}-\mathbf{r}'|} d\tau'$ (localized currents only)

$\oint \mathbf{A} \cdot d\mathbf{l} = \Phi_B$

USEFUL FACTS

A points along J

$A_z(s)\hat{z}$: $B_\phi = -\frac{dA_z}{ds}$

$A_\phi(s)\hat{\phi}$: $B_z = \frac{1}{s}\frac{d}{ds}(sA_\phi)$, or $A_\phi \cdot 2\pi s = \Phi$

$A_x(z)\hat{x}$: $B_y = \frac{dA_x}{dz}$

$A_\phi(r,\theta)$: $B_r = \frac{\partial_\theta(\sin\theta A_\phi)}{r\sin\theta}$, $B_\theta = -\frac{\partial_r(rA_\phi)}{r}$

A continuous; $\partial_n \mathbf{A}$ jumps by $-\mu_0 \mathbf{K}$; finite on axis

Infinite currents: the integral diverges ⇒ solve the curl ODE instead

SPECIFIC RESULTS

10.1 Wire with J = k s^n ẑ, A(0) = 0



$B_\phi = \frac{\mu_0 k s^{n+1}}{n+2}$; $A_z = -\frac{\mu_0 k s^{n+2}}{(n+2)^2}$

J = k s²: $B_\phi = \frac{\mu_0 k s^3}{4}$, $\mathbf{A} = -\frac{\mu_0 k s^4}{16}\hat{z}$

Check: $\frac{1}{s}(sA_z')' = -\mu_0 J$

10.2 Uniform wire, everywhere



With A(a) = 0: in $\frac{\mu_0 J}{4}(a^2 - s^2)$; out $-\frac{\mu_0 I}{2\pi}\ln\frac{s}{a}$

Coax between: $-\frac{\mu_0 I}{2\pi}\ln\frac{s}{b}$; outside 0

10.3 Solenoid / uniform B



Uniform B: $\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r} = \frac{B_0 s}{2}\hat{\phi}$

Solenoid: $\frac{\mu_0 n I s}{2}\hat{\phi}$ in, $\frac{\mu_0 n I R^2}{2s}\hat{\phi}$ out

10.4 Spinning shell, dipole, magnetized sphere



Shell in: $\frac{\mu_0 R\omega\sigma}{3} r\sin\theta\hat{\phi}$; out: $\frac{\mu_0 R^4 \omega\sigma \sin\theta}{3r^2}\hat{\phi}$

Dipole: $\frac{\mu_0 m \sin\theta}{4\pi r^2}\hat{\phi}$; M-sphere in: $\frac{\mu_0 M}{3} r\sin\theta\hat{\phi}$

10.5 Sheet and slab HW 5.29



Sheet $K\hat{x}$: $-\frac{\mu_0 K}{2}|z|\hat{x}$

Slab $J\hat{x}$: in $-\frac{\mu_0 J z^2}{2}$; out $-\mu_0 J a|z| + \frac{\mu_0 J a^2}{2}$

10.6 Given A: find B, then J HW 5.26

$A \to B$ ($\nabla\times$), $B \to J$ ($\nabla\times/\mu_0$)

$A = k\hat{\phi}$: $B = \frac{k}{s}\hat{z}$, $J = \frac{k}{\mu_0 s^2}\hat{\phi}$

$A = ks^3\hat{z}$: $B = -3ks^2\hat{\phi}$, $J = -\frac{9ks}{\mu_0}\hat{z}$

Finite wire: $\frac{\mu_0 I}{4\pi}\ln\frac{z_2+\sqrt{z_2^2+s^2}}{z_1+\sqrt{z_1^2+s^2}}\hat{z}$

11 · MAGNETIC DIPOLES & SPINNING CHARGE

GENERAL SETUP



$\mathbf{m} = I\mathbf{a}$ (RHR: fingers with I, thumb = m)

$\mathbf{B} = \frac{\mu_0 m}{4\pi r^3}(2\cos\theta\hat{r} + \sin\theta\hat{\theta})$

$\mathbf{A} = \frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\hat{r}}{r^2}$

USEFUL FACTS

$\mathbf{N} = \mathbf{m} \times \mathbf{B}$; $U = -\mathbf{m} \cdot \mathbf{B}$; $\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$

No magnetic monopole term

Any spinner: $m = \frac{Q}{2M}L$

Ring ½, disk ¼, shell ⅓, ball ⅕ (× QωR²)

Average B inside a sphere: $\frac{\mu_0}{4\pi}\frac{2\mathbf{m}}{R^3}$

SPECIFIC RESULTS

11.1 Two dipoles HW 6.3



Coaxial: $F = \frac{3\mu_0 m_1 m_2}{2\pi r^4}$ (attract)

$U = \frac{\mu_0}{4\pi r^3}[\mathbf{m}_1 \cdot \mathbf{m}_2 - 3(\mathbf{m}_1 \cdot \hat{r})(\mathbf{m}_2 \cdot \hat{r})]$

11.2 Spinning disk (σ, ω) Disc 5.39



$m = \frac{\pi}{4}\sigma\omega R^4$; center $B = \frac{\mu_0\sigma\omega R}{2}$

Axis: $\frac{\mu_0\sigma\omega}{2}[\frac{R^2+2z^2}{\sqrt{R^2+z^2}} - 2z]$

11.3 Spinning shell (σ, ω) Ex 5.11



Inside $\mathbf{B} = \frac{2}{3}\mu_0\sigma\omega R\hat{z}$ (uniform); $m = \frac{4}{3}\pi\sigma\omega R^4$

Outside: pure dipole

11.4 Spinning solid ball (ρ, ω) Disc 5.63



$m = \frac{1}{5}Q\omega R^2$; center $\frac{\mu_0 Q\omega}{4\pi R}$; average $\frac{\mu_0 Q\omega}{10\pi R}$

$\mathbf{B}_{in} = \mu_0\rho\omega[(\frac{R^2}{3} - \frac{r^2}{5})\cos\theta\hat{r} - (\frac{R^2}{3} - \frac{2r^2}{5})\sin\theta\hat{\theta}]$

11.5 Spinning cylinders



Shell (σ, R, L): $K = \sigma\omega R$, $B = \mu_0\sigma\omega R$ inside, $m = \pi\sigma\omega R^3 L$

Solid (ρ): $B(s) = \frac{\mu_0\rho\omega}{2}(R^2 - s^2)$

12 · B, H, M: MAGNETIC MATTER

GENERAL SETUP



$\mathbf{M}$ = magnetic moment per volume (A/m)

Bound current: $\mathbf{J}_b = \nabla \times \mathbf{M}$, $\mathbf{K}_b = \mathbf{M} \times \hat{n}$

$\mathbf{H} \equiv \frac{\mathbf{B}}{\mu_0} - \mathbf{M}$: $\nabla \times \mathbf{H} = \mathbf{J}_f$, $\oint \mathbf{H} \cdot d\mathbf{l} = I_{f,enc}$

$\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$ (NOT zero in general)

USEFUL FACTS

Linear: $\mathbf{M} = \chi_m \mathbf{H}$ and $\mathbf{B} = \mu\mathbf{H}$

$\mu = \mu_0(1 + \chi_m) = \mu_0\mu_r$, so $\mu_r = 1 + \chi_m$

Homogeneous: $\mathbf{J}_b = \chi_m \mathbf{J}_f$

Para: $\chi_m > 0$ (~10^{-5}), pulled in; dia: $\chi_m < 0$, pushed out; ferro: keeps M

No free current + symmetry ⇒ H = 0 ⇒ B = μ_0M

Free current: H from Ampère, then $\mathbf{B} = \mu\mathbf{H}$ (or $\mu_0(\mathbf{H} + \mathbf{M})$)

Shape: $\mathbf{B}_{in} = \mu_0(1 - N)\mathbf{M}$, $\mathbf{H}_{in} = -N\mathbf{M}$

P charges the caps; M puts current on the sides

SPECIFIC RESULTS

12.1 Uniformly magnetized sphere



$\mathbf{K}_b = M\sin\theta\hat{\phi}$ (= spinning shell with σωR → M)

$\mathbf{B}_{in} = \frac{2}{3}\mu_0\mathbf{M}$, $\mathbf{H}_{in} = -\frac{\mathbf{M}}{3}$

Outside: dipole $\frac{4}{3}\pi R^3\mathbf{M}$; uniform-M shell: B = 0 in cavity

12.2 Magnetized rod / disk



Long rod ∥ M: $B = \mu_0 M$ (solenoid with nI → M)

Thin disk ⊥: B ≈ 0, H ≈ −M

Finite rod on axis: $\frac{\mu_0 M}{2}(\cos\theta_2 - \cos\theta_1)$

12.3 Cylinder with M along φ̂ HW 6.8



$M = ks^2\hat{\phi}$: $\mathbf{J}_b = 3ks\hat{z}$, $\mathbf{K}_b = -kR^2\hat{z}$

$\mathbf{B} = \mu_0\mathbf{M}$ inside, 0 outside (H = 0)

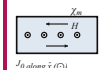
12.4 Cylinder with M = k s ẑ HW 6.12



$\mathbf{J}_b = -k\hat{\phi}$, $\mathbf{K}_b = kR\hat{\phi}$, $\mathbf{B} = \mu_0 ks\hat{z}$ in, 0 out

Add free wire I on axis: $H = \frac{I}{2\pi s}$, $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$

12.5 Linear slab carrying free J_0 x̂, |z|<d/2



$\mathbf{H} = -J_0 z\hat{y}$; $\mathbf{M} = \chi_m\mathbf{H}$; $\mathbf{B} = \mu_0(1 + \chi_m)\mathbf{H}$

Outside: $\mathbf{H} = \mp\frac{J_0 d}{2}\hat{y}$, $\mathbf{B} = \mu_0\mathbf{H}$

$\mathbf{J}_b = \chi_m J_0\hat{x}$; $\mathbf{K}_b = -\frac{\chi_m J_0 d}{2}\hat{x}$ on both faces

Net bound current = 0 ✓

12.6 Coax / toroid / solenoid with χ_m HW 6.16



$H = \frac{I}{2\pi s}$; $B = \frac{\mu_0(1+\chi_m)I}{2\pi s}$; $M = \frac{\chi_m I}{2\pi s}$

$\mathbf{K}_b = +\frac{\chi_m I}{2\pi a}\hat{z}$ inner, $-\frac{\chi_m I}{2\pi b}\hat{z}$ outer; $J_b = 0$

Core in solenoid/toroid: B × (1 + χ_m)

12.7 Linear sphere / slab in uniform B_0

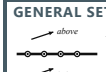


Sphere: $\mathbf{B}_{in} = \frac{3\mu_r}{\mu_r+2}\mathbf{B}_0$

Slab ⊥ B_0: B same, H ÷ μ_r; slab ∥ B_0: H same, B × μ_r

13 · MAGNETIC BOUNDARY CONDITIONS & E ↔ B

GENERAL SETUP



$B^\perp$ continuous; $\mathbf{B}_{above} - \mathbf{B}_{below} = \mu_0\mathbf{K} \times \hat{n}$

$\mathbf{H}^\parallel$ jumps by $\mathbf{K}_f \times \hat{n}$

$\mu_1 H_1^\perp = \mu_2 H_2^\perp$ (no free surface current)

A continuous; $\partial_n \mathbf{A}$ jumps by $-\mu_0\mathbf{K}$

USEFUL FACTS

ρ ↔ J; ϵ_0 ↔ $1/\mu_0$; V ↔ A (per component)

Gauss ↔ Ampère; $\frac{p}{4\pi\epsilon_0}$ ↔ $\frac{\mu_0 m}{4\pi}$

D ↔ H; P ↔ M

Sphere: $-P/3\epsilon_0$ ↔ $+\frac{2}{3}\mu_0 M$; long rod: ≈0 ↔ μ_0M

Fill all space: E ÷ ϵ_r ↔ B × μ_r